

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 2

Hardware and software

59 questions from 28 question papers, Feb/March 2022 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

Questions

9626/12 Feb/March 2022 · Question 2

[Mark scheme](#)

- 2** An interpreter is a type of system software.

Describe how an interpreter operates.

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9626/12 Feb/March 2022 · Question 3

[Mark scheme](#)

- 3** Mainframe computers are particularly suited to carrying out the transaction processing of online banks.

(a) Describe the features of a transaction processing system.

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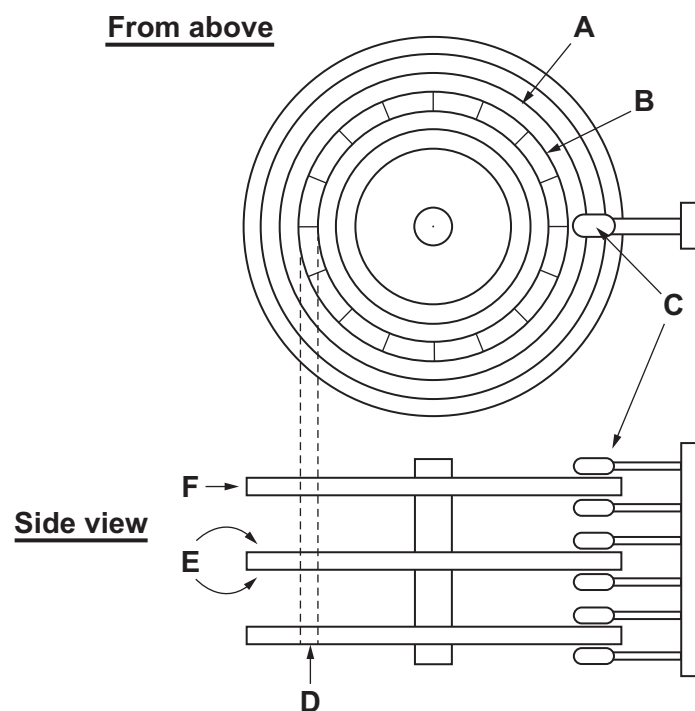
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..... [3]

(b) Discuss the advantages and disadvantages of the use of mainframe computers.

[8]

5 Here is a diagram showing the component parts of a hard disk drive.



- (a) Describe the component parts of a hard disk drive. Refer to the labelled diagram in your answer.

[6]

(b) Explain what is meant by disk defragmentation and why it is needed when using hard disks.

[4]

9626/11 May/June 2022 · Question 11

Mark scheme

11 A gesture based interface is a type of user interface.

Describe the advantages of using a gesture based interface rather than other types of interface.

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- 2 Explain how high level language programs can be translated to run on different computer systems, including any difficulties which may result from this.

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Discuss the advantages and disadvantages of the doctor using a dialogue interface rather than a graphical user interface (GUI).

[8]

1 Computers often have utility software supplied with the operating system.

Explain what is meant by utility software and why it is needed.

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Mark scheme

- 6** Explain why supercomputers are used for weather forecasting.

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- 3** Back-up software and anti-virus software are both types of utilities that are often provided with operating systems.

- (a)** Without detailing the different methods of detection, describe the features of anti-virus software.

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- (b)** Without referring to the different types of back-up, describe the features of back-up software.

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6 The IT department manager in a company has decided new software is needed to run the payroll system. She will make a choice between using off-the-shelf software and custom written software. She is already aware that there are differences in the level of support offered and the amount of testing that will have been carried out on the software.

Discuss the other advantages and disadvantages of choosing off-the-shelf software compared to choosing custom written software.

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3 Most computer systems have a variety of utility software including anti-virus software.

Discuss the benefits and drawbacks of anti-virus software.

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[6]

9626/11 May/June 2023 · Question 3

Mark scheme

3 Explain why a compiler may not be suitable for translating a high level language program for use on different computer systems.

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[8]

3 Companies have the choice of using custom written software or using off-the-shelf software.

Explain the differences between custom written software and off-the-shelf software in terms of:

(a) cost

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(b) level of support

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(c) degree of testing.

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2 Mainframe computers are used in banking. Supercomputers are used in climate research.

(a) Explain what is meant by a mainframe computer and by a supercomputer with reference to the types of processing and the processors they each use.

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(b) Explain how supercomputers are used in climate research.

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3 Give brief descriptions of the following types of user interface.

(a) command line interface
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(b) gesture based interface
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..... [2]

(c) dialogue interface
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- 3 Mainframe computers and supercomputers are specialised types of computer with some similar characteristics.

Explain what is meant by each of the following characteristics.

- (a) fault tolerance

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- (b) heat maintenance

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Mainframe computers were developed from the need for computers to process data in tasks such as carrying out a census.

- (c) Explain why mainframe computers are used to process the data from a census.

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3 Keith wishes to ensure that his computer is at its optimum performance.

Identify and describe **three** different types of utility software he should use.

Utility 1
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Utility 2
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Utility 3
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[6]

[6]

7 The IT manager of RockICT can purchase either off-the-shelf software or custom written software.

(a) Describe **three** advantages for RockICT of purchasing off-the-shelf software.

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(b) Describe **three** advantages for RockICT of purchasing custom written software.

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7 Compilers and linkers are types of system software.

(a) Describe the role of a compiler.

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(b) Describe the role of a linker.

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[8]

1 Utility software is needed to maintain a computer system.

Explain why each of the following tasks is needed and how utility software performs each task:

(a) disk formatting

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(b) data compression

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(c) disk defragmentation

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2 Describe the following characteristics of mainframe computers:

(a) volume of input, output and throughput

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(b) security

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[8]

- 9** Utility software is a type of system software designed to help optimise, manage and maintain the performance and health of computer systems.

(a) Explain how the delete utility works.

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(b) Explain why files can still be recovered even after they have been deleted.

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Explain why the manager would choose off-the-shelf software.

[6]

[8]

- 4 A company finance director could use either off-the-shelf or custom-written software to help manage the company’s finances.

Contrast these two types of software.

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Mark scheme

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8 Interpreters and device drivers are types of system software.

(a) Describe the functions of device drivers.

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(b) Justify the use of an interpreter.

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10 A student is considering defragmenting their hard disk.

(a) Describe the benefits of defragmentation.

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(b) Describe the drawbacks of defragmentation.

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6 Describe what utility software is and why it is needed on a computer system.

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9 A programmer is developing a new piece of software. The programmer has decided to write code that needs to be interpreted so that it will run.

Justify the programmer’s decision.

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9626/11 and 9626/13 Oct/Nov 2025 · Question 11

[Mark scheme](#)

11 Describe **four** characteristics of a supercomputer.

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- 2
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- 3
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- 4
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- [4]

9626/11 and 9626/13 Oct/Nov 2025 · Question 12

[Mark scheme](#)

12 Describe what is meant by the term MIPS.

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- [2]

9626/12 Oct/Nov 2025 · Question 10

[Mark scheme](#)

10 Describe **four** characteristics of a mainframe computer.

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- 2
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- 3
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- 4
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- [4]

9626/12 Oct/Nov 2025 · Question 11

[Mark scheme](#)

11 Describe what is meant by the term FLOPS.

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- [2]

12 Describe how an operating system manages data on a hard disk.

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3 A consultant has suggested that a gesture-based interface could be used as part of a security system on the door of a government office. The office is only used by staff.

(a) Describe how a gesture-based interface can be used to interact with the security system.

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(b) Evaluate the use of a gesture-based interface as part of the security system for this office.

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4 Mainframe computers and supercomputers have differing characteristics.

Using **three** different characteristics, describe differences between mainframe computers and supercomputers.

characteristic 1

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difference

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characteristic 2

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difference

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characteristic 3

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difference

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[6]

[8]

[8]

[6]

[4]

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Mark scheme

Discuss, with the example of weather forecasting, the benefits and drawbacks of using a supercomputer.

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6 Disk defragmentation is a type of utility software.

(a) Describe what is meant by disk defragmentation.

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(b) Explain why disk defragmentation is less likely to be used in modern computers.

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..... [1]

Back-up and file deletion are also types of utility software.

(c) Other than disk defragmentation, back-up and file deletion, identify and describe the specific purpose of **two** types of utility software.

type 1
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type 2
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[4]

Mark schemes

Mark scheme · 9626/12 Feb/March 2022 · Question 2

[Back to question](#)

Question	Answer	Marks
2	Four from: Translates a high-level language program... ...into an intermediate stage Translates a line of source code and then executes that line/statement Translates instructions one at a time Reports on errors as lines of source code are processed An interpreter has to be resident in memory in order for the program to run Only a few lines of the program need to be in memory at any one time	4

Mark scheme · 9626/12 Feb/March 2022 · Question 3

[Back to question](#)

Question	Answer	Marks
3(a)	Three from: A transaction processing system's performance is measured by the number of transactions it can process in a given period of time The system must be continuously available The system must be able to maintain the integrity of the data and overcome hardware or software problems It must also be possible to individually upgrade hardware and software components without shutting down the system There should be controlled access, allowing only authorised users to use the system	3

Question	Answer	Marks
3(b)	Eight from: <i>Advantages</i> They can perform hundreds of MIPS/have faster processing than PCs, laptops and other devices. They have a high value of mean time between failures (MTBF) so usually do not have any time where the system is downPCs/laptops/other computing devices can crash more often They have greater fault tolerance than other systems... ...they can transfer processing from one core to another... ...so hardware and software upgrades can occur while the system is still in operation They can run different operating systems so can handle different types of database They are more secure than most other types of computer as they use complex encryption systems. <i>Disadvantages</i> They are very expensive to buy/installsome organisations are changing to using the cloud rather than buying a mainframe... ...there is a higher cost for the people needed to manage them... ...which is not the case with cloud-based computing The software required to run a mainframe is more expensive to buy than using cloud computing Mainframes need large rooms to accommodate them They generate a lot of heat... ...cooling systems are expensive to buy, install and run Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Mark scheme · 9626/11 May/June 2022 · Question 5

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Question	Answer	Marks
5(a)	Six from: A is a track, B is a sector, C are read-write heads, D is a cylinder, E are surfaces, F is a platter (3 marks for all 6 correct, 2 marks 5 correct, 1 mark for 3 or 4 correct) A hard disk drive consists of several platters which are individual disks Each surface of a platter has its own read/write head The read/write heads move across the platters/surfaces stopping only to read <u>data</u> from or write <u>data</u> to the surface... ... never touching a disk surface each of which is used to store data Each surface is divided into several tracks which are in the same position on each disk The track on the top platter together with the tracks exactly below it, form a cylinder Each track is divided into sectors	6
5(b)	Four from: Organises the contents of the disk into the smallest number of contiguous regions/fragments... ...by moving the data blocks of a file/defragmented files around to bring whole/all the parts of a file together Attempts to create larger regions of free space Some defragmentation utilities try to keep smaller files within a single directory together It is the rearranging of files stored on a disk The movement of the hard drive's read/write heads over different areas of the disk when accessing fragmented files is slower... ...compared to accessing the entire contents of a non-fragmented file sequentially... ...data retrieval is made easier and quicker	4

Mark scheme · 9626/11 May/June 2022 · Question 11

[Back to question](#)

Question	Answer	Marks
11	Five from: Gestures may be a quicker way of initiating a response from a device People who have speech impediment can find a gesture based interface useful rather than dialogue Gestures will still be reliable if there is background noise or user has a cold/strong dialect/accent which would affect a dialogue interface Dialogue interface often requires a training session with user whereas gestures can be taught through manuals Users with physical handicaps may find it easier to make gestures rather than gripping a mouse Don't have to learn so many commands as with a CLI For reasons of hygiene doctors/health workers/restaurant chefs may not be allowed to touch a display or device using a GUI... ...appropriate gestures are a more hygienic way to control the device	5

Question	Answer	Marks
4	Six from: <i>Similarities</i> Both are used to measure performance speeds of computers/processors Both can be unreliable as the complexity of instructions can vary according to which <u>benchmark</u> program/test is used/complex instruction takes longer to process than a simple one <i>Differences</i> MIPS are used with mainframes (M)FLOPS used with supercomputers MIPS can be used with integers (M)FLOPS with real/floating point numbers MIPS measured in millions (M)FLOPS measured in quadrillions and higher MIPs is millions of instructions per second, FLOPs is floating point operations per second (must have both) MIPS used when measuring performance of computers involved in running application software... ... whereas FLOPS used where scientific/complex calculations need to be carried out MIPS do not take into account input/output speeds... ... whereas FLOPS do not take into account CPU's clock speed , bus speed, the amount of RAM available Must have at least one similarity and difference to gain full marks	6

Question	Answer	Marks
5	Eight from: <i>Advantages</i> Once it is compiled a program does not have to be translated again A compiled program can run faster than interpreted software Compiling a program is a much quicker process /translates it faster than interpreting... ... as whole program is translated all in one go /all at onceunlike interpreter translates one line at a time Once the program is already compiled into machine language, there is no translation software that the user has to keep up to date Compilers also improve security for programmers... ...makes it much harder to copy code/machine code is difficult to understand and alter... ...so fewer risks of copyright infringement... ... with an interpreted program, the source code can always be seen so greater risk of software copyright infringement <i>Disadvantages</i> Can only be used on one operating system ... source code can be interpreted on any operating system provided interpreter is present... ...will need to use an interpreter if program is to be shared... It is slow/(more) difficult to debug the program/find errors... ...the whole program needs to be compiled again after an error is corrected... ...error messages are output at the end... ...whereas with an interpreter error messages are output <u>as they are found</u> It uses up more memory than an interpreter... ...as whole program has to be loaded... ...interpreter only requires a few statements of the program to be in memory Unlike an interpreter, does not allow small pieces of code to be tested to make sure they work before continuing with the rest of the program More likely to crash the computer as it is running directly on the CPU Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Mark scheme · 9626/13 May/June 2022 · Question 2

[Back to question](#)

Question	Answer	Marks
2	Four from: An interpreted program will still <u>be in its original source code</u> so it will work on <u>any system</u>... ...but must have an appropriate interpreter Only use high level languages which can be interpreted With languages which are compiled can use a cross compiler... ...so programs can run on a computer with a different operating system/target computer... Cross compilers can be a smaller version of the compiler which is normally used on the host computer/native compiler A problem with using a cross compiler is that the compiled program will no longer run on the computer the program was written on/host computer The cross compiled code can run more slowly than if it had been originally compiled on the target machine A cross compiler produces more errors and mistakes than a native compiler	4

Mark scheme · 9626/13 May/June 2022 · Question 4

[Back to question](#)

Question	Answer	Marks
4(a)	Four from: Software that is specially developed for a specific company/business It is made to accommodate that customer's particular preferences and needs Written by programmers to solve <u>specific problems</u> Owned by the business that commissions it It is made from scratch Support is obtained directly from the creators Development costs have to be paid for by client for whom its written/ commissioning company/business	4
4(b)	Four from: Software that is ready-made/ready for use and already exists/can be used immediately It is available to all businesses and companies/general public/everyone It is owned by the company that created it/users only have a licence Has to be adapted to fit the business that has purchased it Development costs are spread amongst several customers Support is obtained from help desks/call centres/user forums run by the producers of the software	4

Question	Answer	Marks
5	Eight from: <i>Advantages</i> For reasons of hygiene a doctor may not be allowed to touch a display or device... ...speaking into a microphone is a more hygienic way to control the device Dialogue interface allows hands free control / ensures safety when driving... ...whereas GUI would need at least one hand to leave the steering wheel Dialogue interface does not require doctor to be sat in front of computer/device so can operate computer/device remotely/allows doctors to be doing other things/tasks at the same time/multitask Users with physical disability may not be able to use a keyboard or mouse... ...or control their limbs accurately... ...could speak into a microphone to control a device <i>Disadvantages</i> Some background conversation may result in a reaction from the device... ...GUIs more likely to require actions to be deliberate Doctors who have a speech impediment may find a dialogue interface too difficult to use Dialogue interface may not operate reliably if there is background noise... ...commands might be misinterpreted/misheard Dialogue interface may not operate reliably if doctor has a cold/strong dialect/accent... ...while GUI will still be reliable Dialogue interface often requires a training session with user/doctor... ...use of GUI can be taught through manuals Dialogue interface can be more expensive to develop compared with a GUI Dialogue interface is not suitable for safety dependant commands such as 'stop' as it would not be as easily understood as GUI Must have at least two of each to gain full marks Must be a proper discussion to obtain full marks Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion	8

Mark scheme · 9626/11 Oct/Nov 2022 · Question 1

[Back to question](#)

Question	Answer	Marks
1	Three from: • A program that helps perform maintenance (in a computer system) (1) • A program that performs a specific task ... (1) ... usually related to managing system resources (1) • Operating systems can contain a number of different utilities/types (for managing files and storage devices) (1) • Identified suitable example (e.g. file management/storage device) (1) One from: • Generic statement of why utility software is required (1) e.g. • It is needed to keep the computer free from viruses (1) • It is needed to make the files contiguous (1) • It is needed to improve the computer's performance by managing the allocation of computer memory (1)	4

Mark scheme · 9626/11 Oct/Nov 2022 · Question 8

[Back to question](#)

Question	Answer	Marks
8	Six from: • Proprietary software is owned by the copyright holder//not owned by the user (1) • Proprietary software users have to purchase a licence/pay to use it ... (1) ... open source software is usually free to download// includes a request for a donation to charity (1) • There are almost always restrictions on the use of proprietary software ... (1) ... open-source software can be used/accessed/implemented by anyone (1) • The specific terms of use are included in an end-user license agreement (1) • The source code of proprietary software is a secret ... (1) ... open-source software source code is open to inspection (1) • Usually covered by copyright which provides a legal basis for its owner to establish exclusive rights ... (1) ... open-source software code can sometimes be amended without breach of copyright legislation (1)	6

Mark scheme · 9626/12 Oct/Nov 2022 · Question 2

[Back to question](#)

Question	Answer	Marks
2	Five from: - Describes the robustness/durability of mainframe computers (1) Reliability is the chance of the system behaving as intended (1) Accessibility refers to the mainframe being operational at all times//the operating time as a percentage of total time (1) Serviceability means it can be easily fixed within a short period of time (1) - Mainframes are reliable as the system will use alternative methods to carry out tasks (if there is a failure) (1) - The system's software is reliable as it is extensively tested and updates are made quickly to overcome problems (1) - Mean time between failures (MTBF) is quite long for mainframes (1) - Mainframes give months/years between system downtimes (1) - Even when unavailable (due to failure) a mainframe is unavailable for a very short time (1) - Spare components/CPU are often included in mainframes so that if one component/CPU fails another takes over (1) - The mainframe can determine why a failure occurred (1) - This enables hardware and software components to be replaced without affecting the mainframe's operations (1) - Can be serviced while the system is still running (1)	5

Mark scheme · 9626/12 Oct/Nov 2022 · Question 6

[Back to question](#)

Question	Answer	Marks
6	Four from: - A supercomputer is fast/powerful/run at several petaflops enough to carry out the calculations that are required for weather forecasting (1) - A supercomputer can process huge amounts of data// a supercomputer is needed because a huge amount of data is collected (1) - Weather forecasting is based on the use of very complex/scientific computer models (which only supercomputers can handle) (1) - A sophisticated/three-dimensional model of the earth's atmosphere/ocean/space has to be created (which needs the processing power of a supercomputer) (1) (A supercomputer is needed) because a large number of calculations have to be carried out//a supercomputer can do a large number of calculations (1) - Needs to do complex/scientific calculations (1) Max two marks for the data that has to be collated: e.g. - records of previous weather conditions (1) - weather measurements from weather stations (candidate may exemplify) (1) - observations from radar/satellites (1) - information from ships /aircraft (1)	4

Question	Answer	Marks
3(a)	Three from: • Scans the computer to make sure it isn't infected with a virus/to find viruses (1) • Can scan whole disk or individual files (1) • Can be set up to scan automatically//run at a set time (1) • Compares with a database/existing viruses (1) • Detects changes in behaviour of files (1) • Quarantines the infected programs (1) • Reports results of scan to user (1) • It will ask the user whether or not they want to delete the infected programs (1) • Does background scans of downloads and attachments (1) • Automatically update itself (1)	3
3(b)	Three from: • Creates additional copies of files/data/databases/hard disks/ network servers (accept 'duplicates') (1) • Backed up//copy of (e.g.) files are still available in the event of data loss//if computer is corrupted/computer fails/(accidental) deletion/other example of data loss (1) • Can use these copies to restore the original contents in the event of data loss//if computer is corrupted/computer fails/(accidental) deletion/other example of data loss (1) • Asks user to enter type of back-up (1) • Can happen automatically (accept reverse argument) (1) • Asks if you wish to restore the back-up (1) • Asks if you wish to verify the back-up (1) • Asks when back-up is to take place/frequency of backups (1) • Ask which medium the user wishes to store/restore backup (1)	3

Question	Answer	Marks
6	Eight from: Advantages <i>Max six from</i> - Off-the-shelf software is cheaper to buy (1) ... as it is mass produced/cost is spread between customers (1) ... whereas with custom written software it costs more to pay programmers to write code specifically for the task/user's needs ... (1) ... written for one purpose/don't share development costs with other users (1) - Off-the-shelf software is already available//available <u>straight away (or very near equivalent)</u> ... (1) ... whereas with custom written software client would need to wait whilst the software was being developed (from scratch) (1) Disadvantages <i>Max six from</i> - Off-the-shelf may need to be adapted for use which can be difficult ... (1) ... there may be settings within the software that cannot be changed ... (1) ... whereas custom written software is designed specifically for the task/company requirements (1) - Off-the-shelf software may not have all the features//may not meet needs of the/ required by the company ... (1) ... whereas with custom written software the programmer can ensure the software will have all the features that are required (1) - Off-the-shelf software may have unnecessary features ... (1) ... whereas the programmer will make sure that custom written software has <u>only</u> the features required (1) - Off-the-shelf software may not be fully compatible with the company's computer system ... (1) ... whereas the programmer will have taken the computer system into account making sure that the software is compatible with it (1) Max six marks if bullets/list of points Max six marks if no attempt at expansion	8

Question	Answer	Marks
2	Up to Six marks available: Benefits Max 5 • Programs can be written in modules (1) – ...which requires less RAM (1) – ... so saving cost of memory (1) • Whole program and compiler don't need to be in memory at the same time (1) – ...which requires less RAM (1) – ... so saving cost of memory (1) Both extension marks above may be given twice (once per first mark item) where used appropriately. • A number of programmers can be used to write separate modules ... (1) – ...which saves time compared to one person writing the whole code (1) • If there is an error in the code only that module has to be corrected (1) Drawbacks • Variable names can cause problems... (1) – ...the same variable may have been given different variable names in different modules (1) • Documentation has to be more detailed ... (1) – ... so takes longer to write (1) Max. five marks if bullets/list of points	6

Question	Answer	Marks
3	SIX marks available: Benefits (MAX 4) - Protects from computer viruses/malicious code (accept either ‘finds’ or ‘stops new infections’ (1) - Prevents data/files (in files) becoming corrupted/deleted (1) - Can prevent passwords from being changed (1) - Can prevent user’s email account sending out spam (1) - Can prevent computer from crashing (1) - User can select how regularly the anti-virus scan is to take place (1) - User can select which files to scan (1) Quarantines any infected files (removes BOD) (1) Drawbacks (MAX 4) - Anti-virus software can cost//has to be purchased (any awareness of cost or expense) (1) - Only purchase for one year (1) - Anti-virus needs to be kept up to date ...(1) ...because there always new virri to be countered (1) - Anti-virus can only detect a virus if it is listed in the AV database (1) - Anti-virus background scans can slow a computer down//slow down download process (1) - Can give a false sense of security/does not give 100% coverage (1) - Can result in false positives (award as a concept, not necessarily these words) ...(1) ...what appears to be suspicious activity turns out to be harmless (1) - Needs expertise to personalise (1) Accept examples such as – “Can take a long time to customise account/settings”. - Takes up space on the hard drive (1) - Can only delete the whole file that is infected (1) Max. five marks if bullets/list of points	6

Question	Answer	Marks
3	Three from: - A compiled program will only run (on a computer with) the same operating system (1st as it was originally translated on (1) - The (other) computer system it is transferred to may not have the same operating system (1) - Cross compilers can be used but the compiled program will no longer run on the host computer/computer the program was written on (1) - Cross compiler produces more errors and mistakes than a native compiler (1) - The cross compiled code can run slower than if it had been originally compiled on the target machine (1)	3

Question	Answer	Marks
6(a)	Five from: • Manager checks the space on the target drive/location (1st) – to make sure there is sufficient space for the data (that will be moved) (1) • Manager selects the time they want the back-up to take place (1st) – manager can select to carry out the back up immediately... (1) – BECAUSE the back-up takes place at a time convenient to the manager/when the system is quiet (may explain why timing is convenient) (1) • Manager selects type of back-up (1st) – backing up only the specified data/files (1) – BECAUSE may want to back up whole device OR new files (1) • Manager selects how regularly the back-up is to take place (1st) – BECAUSE to minimise data loss (1) • Manager will decide where back-up is to be stored (1st) – BECAUSE to make sure back-up will be secure (1) • Manager selects whether to verify the back-up or not (1st) – BECAUSE to make sure the backup has been completed correctly (1) • Manager chooses whether or not to encrypt the back-up (1st) – BECAUSE adds an extra level of security (1) Must have at least TWO reasons to obtain full marks	5

Question	Answer	Marks
6(b)	MAX three from: • Can be either lossy or lossless. (1) • Lossless compression reduces number of bits (1st) – ...by identifying repeated patterns of data. (1) • No information is lost in <u>lossless</u> compression. (1) • <u>Lossy compression</u> reduces number of bits (1st) – ...by identifying unnecessary information. (1) Effects max three: • Quality of audio is the same when using <u>lossless compression</u> (1) • Quality of audio is lower when using <u>lossy compression</u> (1) • File size is smaller for lossy than lossless (accept converse) (1)	4

Question	Answer	Marks
5	Five from: Four max from: • A device driver is a piece of (system) software (1) • It operates/controls a device attached to the computer (1) • Allows communication between printer and computer (1) • A device driver tells the operating system/application software how to communicate with a hardware device (1) • A printer driver acts as an interface between the operating system/application software and the printer (1) • When a document is to be printed the application tells the printer driver and the printer driver tells the printer (1) • The user appears to have control of the device (1) • Upon installation it detects and identifies the peripheral device (1) • Handles the translation of requests between a device and the computer (1) • Defines where outgoing data must be stored before it can be sent (1) Two max from: • Without the required (printer) driver the printer fails to work (1) • The software used by a computer tends to be created by different companies to those that manufacture printers (1) • Computers (may) use software which uses a different type of language/instruction set to the printer (1) • The printer driver is needed to convert the language/instruction set so that the software is able to communicate with the hardware (1)	5

Question	Answer	Marks
4	Eight from: • Manages/operates/controls computer hardware resources (1) • Manages software resources (1) • Interacts with device drivers (1) • Responds to input devices/recognises inputs (1) • Sends data/instructions to printers/output devices (1) • Manages files/directories (1) • Responds to inputs (1)// interface between the user and the computer (1) • Allocates memory to software (1) • In multi-tasking/multi-programming systems allocates equitable processing time to each task/program (1) • Supplies utilities for managing the computer – (may state or exemplify) (1) • Outputs to hardware//interface between an application program and the computer hardware (1) • Knows the names of each file and (exactly) where they are stored (on the storage media) (1) • It keeps a record of any empty spaces on the medium to know where new files can be stored (1) • Displays error messages to the user if an error occurs (1) • Opens and closes files (on storage devices) (1) • Allows communication between attached hardware (1)	8

Question	Answer	Marks
9	Eight from: Advantages of GBI/disadvantages of GUI • Can be advantageous to disabled users ... (1) ... who maybe cannot control their hands accurately//have limited mobility (1) ... could make gestures using their limbs/eyes (1) • For reasons of hygiene a doctor/food worker may not be allowed to touch a display or device ... (1) ... gestures control the device/ allow them to control devices without touching anything (1) • Gestures may be a quicker way of initiating a response from a device (1) • Some gestures may be unintentional but still initiate a reaction from the device (1) • Gesture based interface does not require person to be sat in front of computer/device so can operate computer/device from a distance (1) • Enhanced gaming experience (1) Disadvantages of GBI/advantages of GUI • GUIs tend to be more accurate than gestures (1) • Certain gestures may not be socially acceptable/user could feel uncomfortable making gestures//reduced privacy (1) • Some gestures may be unintentional but still initiate a reaction from the device//GUIs require actions to be deliberate (1) • Gesture based is less effective when several users/background movement involved (1) • In certain lighting conditions gesture-based interface may misinterpret/fail to register a gesture (1) • GUIs can be used in any lighting/dark situations (1) • Increased danger of injury (1) Max. six marks if bullets/list of points	8

Question	Answer	Marks
3(a)	Two from: - Companies have to pay the programmers to write the programs which could cost more (than buying off-the-shelf software) (1) - Off-the-shelf software is cheaper to buy because it is mass-produced (1) - With custom written software one company pays for the development because they commissioned it (1st) - with off-the shelf software development costs are spread amongst several customers because it is written for a mass market/generic software (1)	2
3(b)	Two from: - With custom written software there may be only one programmer/small team of programmers so companies find it difficult to get support (1) - With custom written software, help can be obtained directly from programmers so can be more helpful (1) - Help desks/call centres/user forums websites for customers are readily available because software is generic//used by a mass audience (1)	2
3(c)	Two from: - Custom written software may not have been tested as thoroughly/rigorously as off-the-shelf software as fewer users/shorter lead time (1) - The tests that are carried out on custom written software are those which the programmer thinks are necessary based on how they think the software will be used which not necessarily be how it is eventually used (1) - Off-the-shelf software will have been used many times before companies buy it so any bugs will have been identified and fixed (1)	2

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Question	Answer	Marks
12	Eight from: • Have a limited lifespan of just a few years (1st) • and will need replacing/repairing (1) • Carries out complex calculations (1) • Can use complex mathematical formulae (1) • Can handle a large amount of items of data at the same time (1) • Are the fastest data processing computers (1) • Are the most expensive computers to buy/install (1) • Each/processor core(s) each have their own operating system (1) • Use massively parallel processing (1) • Are more powerful compared to mainframes (1st) • can be up to a 1000 times faster (1) • Can include scientific rules (1) • Some tasks do not require super computers (1st) • as the task is not complex, so it would be a waste of e.g. resources (1) • Very likely to overheat (1) (and) difficult to keep cool (1) Max. six marks if bullets/list of points	8

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Question	Answer	Marks
2(a)	MAX two from: • A mainframe computer can have hundreds of (processor) cores (1) • All cores share//has one Operating System (1) • Mainframes use parallel processing (1) • Can process a large number of small tasks at the same time (1) MAX two from: • Supercomputers can have more than 100 000 (processing) cores (1) • Supercomputers use massively parallel processing (1) • Each/Processor core(s) each has their own operating system (1) • Supercomputers use more than one GPU (1)	3

Question	Answer	Marks
2(b)	Four from: - Many (types of) variables are collected (1) - Complex calculations carried out (1) - Identifies TWO items measured by sensors for this purpose (1) - Large quantity of data is collected/handled/processed (for each variable) (1) - Shows the (historical) trends of different variables over several decades/long period of time (1) (Complex) mathematical formulae are used to describe how the different parts of a climate system interact (1) - Computer models are used to describe how the climate will change (1) - Supercomputer models will include several (scientific) rules (1) Model include so many uncertainties//are so complex that there is no known exact solution//answer is a “best guess” (1)	4

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Question	Answer	Marks
3(a)	Two from: - User types in text/instructions//command must be typed in//is a text based interface (1) - Prompt appears in screen (after which command is typed in) (1) - Requires fewer resources than GUI (1) - Requires a keyboard and screen (1)	2
3(b)	Two from: - User moves body/identified body part (to communicate with computer//give instructions) (1) - Requires a camera/sensors (to detect the movements) (1) - Computer matches body movements it has stored (in a database) (1) (Stored) body movements relate to (specific) commands (1) - One mark for any identified body movement (1) e.g. finger pointing, head nodding, hand waving/raising, eyes rolling/blinking	2
3(c)	Two from: - User uses their voice/spoken word (to communicate with the computer/give instructions) (1) - Speech recognition software is used to recognise speech (1) - Computer matches speech it has stored (in a database) (1) - Speech is converted into commands//example of command and action that follows (1) - Requires a microphone (1)	2

Question	Answer	Marks
3(a)	Three from: - Whether a computer will repair itself if an error occurs (1) - Whether a computer can continue to operate even if one (or more) of its components has failed (1st) - without affecting the operation of the system/without downtime (award once only) (1) - Whether a computer's operating quality can reduce, but does not fail completely (1) - Whether when a hardware component/processor fails, another takes over/elements are hot swappable (1st) - without affecting the operation of the system/without downtime (award once only) (1) - Whether computers have two different instances/installations/versions of software (1st) - if there is a problem with the first instance/installation/version, the other version is automatically run (1) - this (characteristic) does not apply to mainframes (1)	3
3(b)	Two from: - Computers generate heat/become overheated (1). - When a computer overheats, it will address this issue (1). - Overheating of mainframes and supercomputers is a problem (1st) - due to the quality and quantity of processors generating a lot of heat (1) - hence the need for cooling systems (1) - Air cooling systems have been used but uses up a lot of power/are very expensive to build and operate (1) - Water cooling systems are more cost-effective/more efficient (1) - in removing any excess heat (1) - Many data centres are now being built in cooler parts of the world (1)	2
3(c)	Three from: - The amount of data that has to be processed is very large (1st) as countries tend to have a lot of people (1) - The amount of data is increasing (1st) as each census produces more data (to add to what already exists)/world populations are rising (1) - All data needs to be processed in a short period of time (1st) - and mainframes have a high speed of processing data (1) (As more data is added), more powerful machines are needed (1st) - and mainframes are easily scalable (1) - Mainframes are very reliable (1st) MAX two marks if answers are all only first marks.	3

Question	Answer	Marks
10	Eight from: • (This) custom written software is created for a defined/this purpose/running a payroll/commissioned by an individual company/business (1) • Does not need to be adapted for use (1) • (Any) settings built into the software can be changed (1st) • if the software doesn't initially meet the manager's requirements (1) • such as ability to copy software on to several devices (1) • Software will have/Programmer can ensure the software will have all the features that are required (1) • Software will be/Programmer will ensure that it is compatible with the existing system and software (1) • Custom written software will not have any unnecessary/distracting features/bloat ware (1) • The company will own the copyright of the custom written software (1st) • so can sell it to others (1) • Usually takes up less storage space than off-the-shelf software (1st) • so less expensive media/data storage costs (1) Max 6 marks for just bullet points	8

Question	Answer	Marks
3	Max six overall. Max two for each suitable utility software. Description may be what the utility does, or how this is achieved, or the impact on optimum performance. e.g. Device driver (1st) e.g. ...allows computer to interact with a device (1) ...manages hardware components/peripherals (1) Data compression tools (1st) e.g. ...to free up storage space (by compressing files)//reduce file size (1) Anti-virus/anti-malware (1st) e.g. ...checks for/DETECTS malicious software (1) ...prevents viruses from being downloaded (1) ...quarantines any virus it finds (1) ...Stops your computer from being infected by a virus (1) File management tools (do not accept examples as first mark) (1st) e.g. ...to organise directories (1) ...delete unnecessary/unused files (1) Searching (1st) e.g. ...to find file duplication (1) ...to find files by name or another criterion (1) (Disk) defragmentation (1st) e.g. ...reassemble files into (contiguous) clusters (1) ...files are able to be opened more quickly (1) Create/resize (logical) volumes (1st) e.g. ...the organisation of disk structure (1)	6

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Question	Answer	Marks
6	Six from: • A CLI is one where the user interacts by typing text commands • Users with physical disabilities may find CLI hard to use (1) as not able to use a keyboard (1) • Instructions have to be learned/memorised • CLI commands are complicated • CLI commands need specific structure • If the syntax is incorrect there maybe unforeseen circumstances//will not work • Increases possibility of human error • A slower way of initiating a response from a device • CLI commands may be processed more quickly (once typed in) • CLI uses less memory than other interfaces • Requires specific training • Does not allow multiple windows/multitasking • More flexible	6

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Question	Answer	Marks
7(a)	Three from: • The software is available now//any awareness that software is more readily available • They can choose between different manufacturers of software • The software is more likely to have been fully tested • Less likely to have errors in the code • More support available from books/manuals/ company staff/user groups • The development cost of the software is spread between multiple customers • Off-the-shelf software tends to be cheaper than custom written software • Updates/patches to the software can be more frequent/often/regular	3
7(b)	Three from: • The software is more likely to be written to work with the existing software • The software is more likely to be written to work with the existing hardware • Can get (direct) support from the developer • The evolution of the software meets changing circumstances within RockICT • Can be written so that less computer resources are used • Only includes the required features • RockICT will own the software and could legally sell it on to others • RockICT has more control of the writing of the software • RockICT will be able to specify exactly what it wants/fulfil the exact purpose for which it has been created.	3

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Question	Answer	Marks
7(a)	Three from: • Translates the whole program as one complete unit/at once (1) • Translates the program into machine code/language (1) • Creates an executable file (1) • Produces a report of errors in the code after compilation (1) • Can optimise source code (to run as fast or as efficiently as possible) (1).	3
7(b)	Three from: • A linker combines object files/modules (1st) – that have been created using a compiler (1) – into an executable file (1) • A linker can link modules from a common library file into the executable file (1) • Replaces symbolic addresses with real addresses (1).	3

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Question	Answer	Marks
2	Six from: • Complex models are created//uses complex calculations (1) • Data is gathered/collects/used/records by sensors (1st) – Identifies one suitable sensor//variable gathered (1) • Vast amount of data is captured/collected (1) • Data is input into the model (1) • Vast number of calculations carried out (1) • Data is collected from around the globe (1) • Searches for a match in past conditions (1) • The software/computer makes prediction(s) – based on past data/past patterns/past conditions (1) – (and) based on current data/current patterns/current conditions (1) • Selects most likely prediction (from a range of possibilities) (1) • Creates output charts (to screens) (1) • Uses plotters to print out charts/maps of pressures/wind speeds. (1)	6

Question	Answer	Marks
9	Eight from: Advantages MAX SIX: e.g. • Users can use speech to interact with the computer/do not need to learn the gestures (1) • It is hands free (1st) – can be used in vehicles/telephone systems (1) ○ is safer than gesture based (ONLY award when this fits the given context in the answer) (1) • More flexible, can use many different commands (1) • Do not need direct line of sight/close proximity (1) • Works in low visibility/ Not affected by background movement (1) • Useful for people with sight loss (1) • Useful for people with mobility issues//disabilities (1) • Does not involve unacceptable gestures (1) Disadvantages MAX SIX: e.g. • Potential loss of privacy (1st) – always listening (1) • Verbal input can be heard by others (1st) – So exposes private information (1) • If they are too close together they can respond to unintended commands (1) • Requires ability to speak//speak clearly speech to be able to follow instruction (1) • Needs to learn the user's speech pattern (1) • Could have problems if there is any background noise (1) • Not suitable for safety/critical instructions (1) • Need to use commands in a certain format/pattern (1) • Can be language dependent//impacted by a dialect//person may speak unsupported language (1)	8

Question	Answer	Marks
1(a)	THREE marks available Max TWO marks: - To prepare a data storage medium for initial use (1) - The purpose of low-level formatting is to prepare the disk surface to receive data/allow the user to save data/files (1) - High-level formatting creates a file system on the disk (1) - Allows the operating system to use the disk space to store and access files (1). Max TWO marks: - Low-level formatting divides the disk surface into tracks/sectors/cylinders (1) - Write heads magnetise the disk areas (1) - Tracks are numbered starting from 0 (1) - A gap is left between tracks (1) - Each track is organised into numbered sectors, starting at 1 and separated by gaps (1)	3
1(b)	THREE marks available: - Identifies Lossless and identifies Lossy compression (1) Max TWO marks - Reduces the overall number of bits/bytes (in a file)//reduces file size (1) - To increase available storage space on a hard drive (1) - File compression programs remove redundancy (1) Max TWO Marks - A lossless file compression program uses codes to represent repeated Information (1) - Lossless compression removes meta data (1)	3
1(c)	THREE marks available: Max TWO marks: - It speeds up the access time to a file (1). - To create larger regions of (contiguous) free space (1) Max TWO marks: - Organises the contents of the disk into the smallest number of contiguous regions/fragments (1) - The data blocks of a file/defragmented files are moved around to bring whole/all the parts of a file together (1)	3

Question	Answer	Marks
2(a)	THREE from: • Mainframes have peripheral processors that deal with all input and output operations (1st) • this leaves the CPU to concentrate on the processing of data (1) Volume of Input (Max 1) • Mainframe computers deal with a very high level of input (1) Volume of Output (Max 1) • Mainframe computers deal with a very high level of output (1) Throughput (Max 1) • This refers to how many processes can be carried out per day (1) • Mainframes can carry out many transactions per day//have a fast rate of processing (1) • Throughput is a discrete mainframe characteristic. (1)	3
2(b)	THREE from: • Mainframe computers are more secure than other types of computer systems (1) • Mainframe has many layers of security (1) • Uses different levels of access (1st) • depends on a user's level of security as to which sets of data they can access (1) • Data that is transmitted and data within the system are encrypted//mainframe computers use encryption to protect data (1) • System is continually monitored for unauthorised access attempts (1)	3

Question	Answer	Marks
7	EIGHT marks available e.g. Benefits max 6 • Compilation process only needs to be done once (1) • Once the program is already compiled into machine language, there is no translation software that the user has to keep up to date (1) • Compiled software can run faster than interpreted software (1) • Compiling a program is a much quicker process than interpreting the code (1) • Compilers compile the code all at once/interpreters converts the code line by line (1) • Do not need to share source code (1) • Compilers also improve security for programmers (1st) by making it much harder to copy the code (1) Drawbacks max 6 • Can only be used on the operating system it was compiled upon (1) • Only produces error list once whole process has been completed (1) • It is slow/more difficult to debug the program (1) • The whole program needs to be compiled again after an error is corrected (1) • It uses up more memory than an interpreter as the whole program has to be loaded (1) • Once compiled, any changes to the original high-level code will require the program to be compiled again (1).	8

Question	Answer	Marks
9(a)	One mark per bullet point to a maximum of three marks. • Does not actually erase/delete the data (1). • Removes the reference from the directory table//index in the HDD/SSD. (1st) – so that the file can not be found//search facility would not work (1) • Data is still written/remains on the storage device (1st) – until overwritten by new information. (1)	3
9(b)	Two marks for a full explanation. • Because data has not been removed/deleted from storage device (1) instead space it has been made available for the OS. (1) • Because while no new information overwrites old/previous/deleted data (1) it can potentially be recovered/accessed using specialised software. (1) • Because they are sent to the recycle bin (1)	2

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Question	Answer	Marks
5	One mark per bullet point to a maximum of six marks. • Can be available quicker than custom written software//is readily available (1) • Costs to buy are less than custom written software (1) • Has been fully tested before launch (1) • More (informal) support available (1st) – Example of support other than direct support from the company (1) • Company provides a help desk (1) • May offer more features (1) • Increased flexibility//not task specific (1)	6

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Question	Answer	Marks
11	Max 6 marks: <i>Benefits</i> • Program code will run on different operating systems (1) • Only need to maintain one version of the program code/source code (1) • Less time will be spent on program code maintenance (1st) – When updates are needed (1) • Errors are highlighted/user is alerted as they are encountered (1) it is easy to debug the program (1) • As only a few lines of program code are in memory (1) uses less memory than a compiler (1) • Code can run in a virtual mode (1st) – so is less likely to crash the computer (1) Max 6 marks: <i>Drawbacks</i> • The interpreter software needs to be kept up to date (1) • If the interpreter is updated then the program code may need to be changed to work with it (1) • Code has to be interpreted every time it is run (1) • Interpreted programs run slower than compiled programs (1) • It stops every time an error is encountered (1st) – interpreting a program is a much slower process than compiling (1) • The source code must always be shared (1) • Source code is easier to convert by fraudulent users (1) • Easier to copy the program. (1)	8

Question	Answer	Marks
4	One mark per heading to a maximum of four marks. Cost: valid comment on cost e.g. • Custom written software can be more expensive (than off-the-shelf software) (1) Ownership: valid comment on ownership e.g. • Custom written software is owned by the company that commissions/creates/buys it (1) • off-the-shelf software means the company only has a licence (1) Degree of testing: valid comment on degree of testing e.g. • Custom written software is tested by the users as part of its development (1) Level of support: valid comment on level of support e.g. • Custom written software means that support is available directly from the creators (1) • Off-the-shelf usually provide call centres which may not be able to solve problems specific to the company (1) Customisation: valid comment on customisation e.g. • Off-the-shelf software may have several unnecessary features that the company may not need/custom written software has only the features the user needs (1) • Off-the-shelf software has to be adapted to fit the company that has purchased it (1) Availability: valid comment on availability e.g. • Off-the-shelf software is available to all businesses and companies/general public not just a specific company/business (1) • Off-the-shelf software is ready-written/ready for use and already exists/there is often a delay between commissioning and receipt of custom written software. (1) Compatability: valid comment on compatabilty e.g. • Off-the-shelf software may not work with your current hardware (1)	4

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Question	Answer	Marks
10	Six marks from: - No need to physically interact (may exemplify) (1st) - Can be more hygienic (1) - Can undertake other tasks at the same time (1) - Can operate the device remotely (1) - Can be triggered without meaning to (1) - More accessible to disabled users (may refer to device itself or devices around the home) (1) - The dialogue interface relies on speech/sound (1st) - User has to be in range (1) - Can undertake other tasks at the same time (1) - Can operate the device remotely (1) - External noise may result in the dialogue interface not understanding the command (1) - If a person has a speech impediment, may find dialogue interface difficult to use (1) - Dialogue interface may misinterpret what is said (1) - More accessible to disabled users (may refer to device itself or devices around the home) (1) - Person may not speak supported language/dialect (1) - Audible acknowledgement (1st) - Know that instruction/command has been actioned (1) - One mark for a valid conclusion (1)	6

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Question	Answer	Marks
12	One mark per explanation to a maximum of three marks. Banks require 24/7 operation and mainframe computers have a high level of Reliability Availability and Serviceability (RAS) to keep the system running (1) Banks require 24/7 operation - ONE FROM - and it must be possible to upgrade/replace/maintain the system without shutting down the system (1) - Mainframe Computers have a high level of serviceability//avoid downtime (1) Banking transactions need to be highly secure and mainframe computers have high levels of security (1) Banks hold sensitive information so high fault tolerance of mainframe computers means that data integrity is maintained (1) - Mainframes have the ability to transfer processing from one core to another so that processing is uninterrupted if it breaks down and so bank service is not interrupted//needs to be 24/7 (1)	3

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Question	Answer	Marks
8(a)	One mark per bullet point to a maximum of four marks. • It operates/controls a device attached to the computer (1) • It provides an interface between operating systems/applications software and the device (1st) – Which handles the translation of requests between a device and the operating system (1) • It allows an operating system to detect/identify the peripheral device (1) • Activates the device when it is needed and deactivates it when it is not needed (1) • Logs and reports errors (1) • Defines where data must be cached (1)	4

Question	Answer	Marks
8(b)	One mark per bullet point to a maximum of four marks. • Translates a high-level language into machine code (1) • Creates an executable file (1) • Identifies errors in code (1) • Stops interpreting code when an error is found//shows errors as it translates (1) • Errors are dealt with as they are found//immediately (1) • Uses less RAM as only one line at a time is stored/processed (1)	4

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Question	Answer	Marks
10(a)	One mark per bullet point to a maximum of three marks. • Optimises space on the disk (1) • Files are better organised (1) • Increases reading and writing speeds (1) • Related files are kept together (1) • Files are contiguous (1) • Data retrieval is quicker//files are opened more quickly (1) • Reduces the movement of the read/write heads (1) • Increases longevity/life span of the disk. (1)	3
10(b)	One mark per bullet point to a maximum of three marks. • It can take time to do (1) • It can use a lot of computer memory (1) • It can use a lot of computer processing (1) • It can make the computer less responsive (1) • It can lead to file corruption (1) • The power must be on/uninterrupted throughout the whole process. (1) • Shortens the life of SSDs (1)	3

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Question	Answer	Marks
6	Max one for: - It is a type of system software/Software used by the operating system (1) Max two from: - Used by the operating system to maintain a computer/file system/add functionality (1) - It improve/optimises the computer's performance (1) - Without utility software some devices/peripheral devices would not be available/accessible to the operating system (1) - Describing a clear specific need that is solved by a named/example of utility software (1)	3

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Question	Answer	Marks
9	Four from: - Programmer can write in high level language (1) - Programmer does not need to code in machine code (1) - The program code will run on different computers/operating systems (1) - Only need to maintain one version of the program (1) - It is easier to debug the program//It is easier for the programmer to spot mistakes in the code (1) - An interpreter will highlight mistakes as they are found (1)	4

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Question	Answer	Marks
11	Four from: - They have a short lifespan (1) - They perform consistently and reliably (concept of reliability) (1) - They have very little downtime (concept of availability) (1) - They are no more complex to maintain than other computer systems (concept of serviceability) (1) - They can sacrifice security for faster processing//they might be more liable to attacks because of the data they contain (concept of increased security) (1) - They have high fault tolerance (1) - They achieve very high temperatures//They require heat maintenance (1) - They have a very large number of processors/multi-core processors running in parallel (1) - Their processors are close together (in a 'computer cluster') (1) - They can run multiple operating systems at the same time (1) - They carry out multiple processes at the same time (1)	4

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Question	Answer	Marks
12	Two from: - Stands for millions of instructions per second (1) - Are a unit of measurement of processing power (1) - Are used to measure mainframe performance (1)	2

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Question	Answer	Marks
10	Four from: - They have a long lifespan (1) - They perform consistently (1) - They have very little downtime (1) - They are easier to maintain/hot swap (1) - They have stronger security (1) - They have high fault tolerance (1) - They achieve very high temperatures/They require heat maintenance (1) - They have a large number of processors/multi-core processors/uses parallel processing (1) - They can carry out multiple processes at the same time - They can run multiple operating systems at the same time (1) - They have a high degree of backwards compatibility (1)	4

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Question	Answer	Marks
11	Two from: - FLOPS is floating point operations per second (1) - Are a unit of measurement of processing power (1) - Are used to measure supercomputer's performance (1)	2

Question	Answer	Marks
12	Four from: Organisation e.g. • It organises the arrangement of the files on the disk (1st) Max one from: – uses indexes to determine where to put data (1) – organises data by (two from) cylinder, head, sector (1) – organises data by memory location/index (1) Maintenance e.g. • Maintains / creates directory structures (1) • Random access/serial access to files (1) • Disk formatting tools/Defragmentation tools (any mention) (1) Access e.g. • Controls access / implements access rights / implements password protection / Makes file sharing possible (1) • Specifies tasks that can be performed on a file (two from e.g. open, close, delete, copy, create, move) (1)	4

Question	Answer	Marks
3(a)	TWO marks available: • (Employee/user) moves part of body (may state part e.g. head/wave of hand) (DO NOT accept 'gestures are made') (1) • Movements are captured by a camera (1) • (Movements/gestures are) compared to a database (of stored movements gestures) (1st) Associated commands are triggered/enacted (may exemplify) (1)	2
3(b)	FOUR marks available: Positives Max 3 • No need to have a pass card/word/code (1st) So fewer things to lose//can't get in by finding a card (1) • No need for physical contact//operate from a distance (1st) Therefore reduced spread of diseases (1) Negatives Max 3 • Gestures may be seen from a distance and copied by others (1st) Therefore reduces the security of the system (1) • Gestures may take time to complete (1st) Therefore are slower//impact of being slower (security risk if trying to avoid someone etc.) (1) • Gestures may be misinterpreted//possibility for false positive//high degree of accuracy required (1st) So incorrect action occurs (accept any realistic action) (1) • System may be inaccessible to disabled staff who cannot make the gesture//if carrying something (1st) Therefore they can not get into the office (1) • Only works when visible to the camera (1st) Delays ability to open door (1) • Gestures may be embarrassing/offensive (1) Conclusion (may only be awarded if argument(s) for and/or against are included) • One mark for a value judgement/conclusion (1)	4

Question	Answer	Marks
4	SIX marks available: ONE mark per identified characteristic, ONE mark for the difference: • Longevity (1st) Mainframe computers last longer than supercomputers (1) • RAS/Reliability, availability and serviceability (1st) Mainframe computers are more reliable/available/serviceable than supercomputers (1) The term RAS is NOT used to assess supercomputers but does refer to mainframe computers (1) • Security (1st) Mainframe computers are more secure than supercomputers (1) • Performance metrics//any statement of processing (1st) Supercomputers are measured in FLOPS/TEPS and mainframes are measured in MIPS (1) Supercomputers are more powerful computers than mainframe computers (1) • Volume of input, output and throughput (1st) A mainframe has a higher rate of throughput//can deal with a large number of simultaneous transactions (1) • Fault tolerance (1st) Mainframe computers have a high level of fault tolerance (1) Mainframe computers allow hot swapping (1) Mainframes have a high mean time between failures (MTBF) (1) • Operating system (1st) Super computers tend to only use Linux (derivative) operating systems (1) Mainframe computers can have multiple operating systems (1) • Number of processors (1st) Mainframe computers have multiple processors (1) • Cores (1st) Supercomputers have more cores than mainframe computers (1)	6

Question	Answer	Marks
9	EIGHT marks available: Areas for consideration (one mark each) • Ability to dictate how the product is tested//Extent to which testing has occurred//Potential for existence of bugs (1) • Extent to which user is/can be involved in testing (1) • Extent to which software has features as required//meets the requirements//tailored for the business (1) • Extent to which software includes unnecessary features//ONLY required features (1) • Compatibility with current system (1) • Copyright ownership//ability to sell on (1) • Who pays/paid development costs (1) • Whether immediately available for use or not (1) • Availability of support (online forums etc.) (1) • Updates (costs of or availability of) (1) • Reliance on continued existence of vendor going forward//product ceases to be available (1) • Scalability (1) Marks may only be awarded where the point made is correctly linked to the type of software. MAX six marks where only custom-written or off-the-shelf are considered.	8

Question	Answer	Marks	Guidance
9(a)	SIX marks available: Effect of icons • Avoids the need to read (1) • Graphical icons engage on an emotional level//are more appealing//visually appealing (1st) Therefore, more likely to be completed (1) Results are more reliable/accurate (1) • Less time is required to record customer satisfaction (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) • Graphical icons are simple to understand (1st) Therefore people will not need to queue/stop to be able to read them (1) Results are more reliable/accurate (1) • Graphical icons are universal/international/not language dependant (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) • Icons will be more widely understood (1st) Therefore more likely to respond (1) Results are more reliable/accurate (1) Languages • No need to use a 'select language button' (1st) Therefore people will not need to queue/stop to be able to read them (1) Therefore more likely to respond (1) • No guarantee that everyone speaks the same language//the language of the country in which the airport is (1) • In an airport, there will be a range of languages spoken (1)	6	Only award answers that support the use of a GUI as a method Do not award • repetition of explanations • answers about the system finding icons easier to process//.impact of use of icons on processing etc. • Users of the airport may be in a rush • GUI more easy to understand • Easier to navigate/use

Question	Answer	Marks	Guidance
9(b)	FOUR marks available: - Lots of noise (may interfere with vocal/vocal may not be clear) (1st) Therefore the computer may misinterpret what has been said (1) - Can not be used by certain disabilities (1) because these users cannot (e.g.) speak (1) - Would need to support all/many languages//may not support a specific languages (1st) Would need a very large database of languages/dialects (1)	4	Mark first two answers only. Do Not Accept: Will have difficulty with accent

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Question	Answer	Marks	Guidance
9(a)	ONE mark available: Software that helps to manage/maintain/optimise a computer system (1)	1	
9(b)	ONE mark available: - Back-up software (1) - Encryption software (1) - Disk checking software (1)	1	

Question	Answer	Marks	Guidance
5	SIX from: MAX 5 if no reference to weather forecasting Benefits (Max four) • Very fast processing//has a high FLOPS (1) • Weather forecast can be produced quickly (1) • Can handle a great deal of information (NOT store) (1) • Produce more/very detailed forecasts (1) • Consider a wide range of possible scenarios (1) • Decision is data driven (1) Drawbacks (Max four) • Supercomputers struggle to predict extreme events//working with extreme data (1) • Lacks human intuition (1) • Supercomputers have a limited lifespan (1) • Supercomputers are expensive to buy (1) • Supercomputers require specialist maintenance teams (1) • Supercomputers are very large//take up a lot of space (1) • Supercomputers require a lot of heat management//generates a lot of heat (1)	6	Do Not Accept: • ...and supercomputers can do this Some marks are awarded based on the use of <u>superlatives</u>. Candidates MUST have a clear awareness of the absolute vastness of the processing speed, or of the amount of information able to be handled. Therefore, if this is established, words such as 'a lot' or 'a great deal' can be awarded.

Question	Answer	Marks	Guidance
6(a)	ONE from: - The rearrangement/reorganisation of data/files (on a storage medium) to ONE OF (1) - create contiguous files - ensure efficient storage/access	1	Do Not Award Move or put together
6(b)	ONE from: - Modern computers use SSD storage (1) - SSD storage does not spin/is static (1) - SSD storage does not fragment files (1) - Defragmentation of an SSD reduces its (effective) life (1)	1	

Question	Answer	Marks	Guidance
6(c)	FOUR marks available (TWO per fully described type of utility software): e.g. • Anti-Virus (1st) Detect/remove viruses/malware (1) • Data compression (1st) To create extra storage/decrease the file size of files (1) • File copying (1st) To make another/ a second version of a file (1) (do not accept 'to copy the file') • Disk formatting (1st) To prepare a disk for first use//to return to disk to factory settings (1) Verify all tracks/sectors are ready for recording data (1) • Disk cleaners (1st) Remove unwanted files from the hard disk (1) • System optimisers (1st) Manage system resources so that computer's performance is improved (1) • Disk encryption (1st) Converts data on the disk into an unusable format (1) • Firewall (1st) Monitors/controls network traffic (1) Acts as a barrier between the network and the outside world (1)	4	Do Not Accept: • Back-up//recovery • File deletion • Disk defragmentation